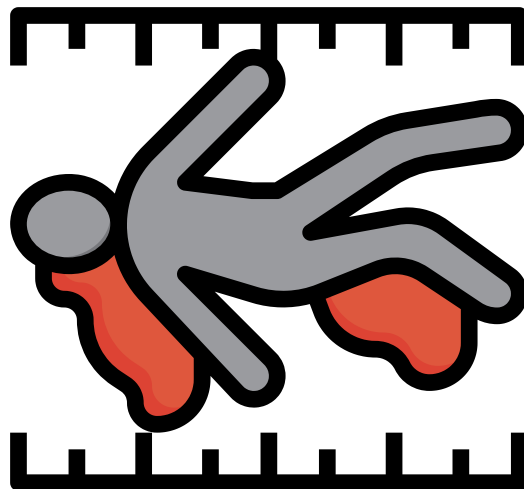
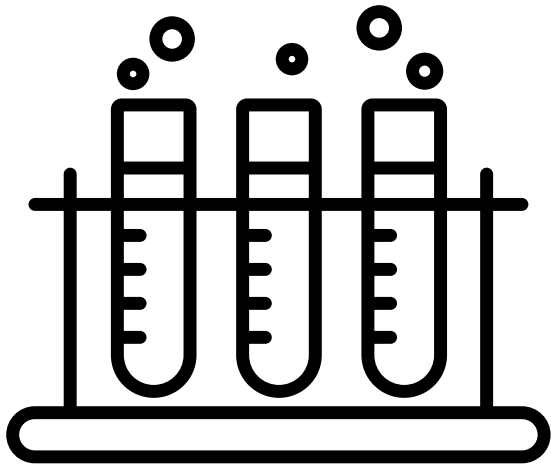


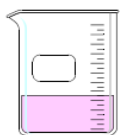
Unit 1 Biochemistry

LAB PACKET

Name: _____



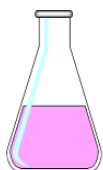
Lab Equipment and Procedures Overview



Beakers are useful as a reaction container or to hold liquid or solid samples. They are also used to catch liquids from titrations and filtrates from filtering operations.



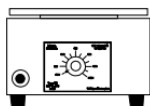
Droppers are for addition of liquids drop by drop



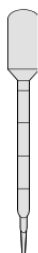
Erlenmeyer Flasks are useful to contain reactions or to hold liquid samples. They are also useful to catch filtrates.



Graduated Cylinders are for measurement of an amount of liquid. The volume of liquid can be estimated to the nearest 0.1 mL with practice.



Hot Plates can also be used as sources of heat when an open flame is not desirable.

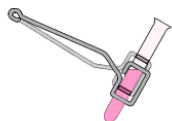


Pipets are used to dispense small quantities of liquids.



reactions.

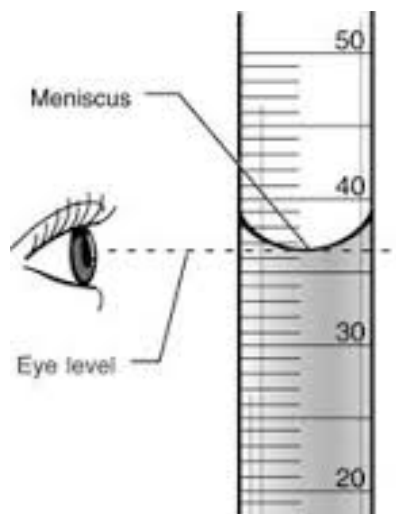
Test Tubes are for holding small samples or for containing



Test tube holders are for holding test tubes when tubes should not be touched

Reading a Meniscus

In all volumetric glassware (pipet, buret, volumetric flasks, graduated cylinder, etc.), it is necessary to read the level of a liquid. A liquid in a small-diameter container will form a **meniscus** or curve at the surface of the liquid. Usually this meniscus curves downward to a minimum at the center. To read the level of the liquid properly, the eye should be at the same level as the bottom of the meniscus.



Using a Digital Scale

1. Open the scale and turn it on.
2. Place a container on the scale- never put chemicals or liquid directly on the scale.
3. Make sure the units are correct. Most often we use grams (g). If it is not correct, press the unit button until it shows the correct units.
4. MOST IMPORTANT STEP: Press the **Tare** button before adding anything else to the scale to zero the scale out.
5. Add your sample or object to the container on the scale.
6. After removing your sample or object make sure you hit the tare button to zero it out again.

Wasp Sting Water Inquiry Lab

Background/Scenario:

You are a field scientist, working with a small group of other scientists to study medicinal plants from the forests of Japan. You work for Georgia Southern University in the Botany department. It is a beautiful day, and you are collecting samples from the Aokigahara forest, when one of the team members lets out a loud scream.

You look over, and see that your colleague is grabbing his/her arm. They have been stung by an Asian Giant Hornet, and they are in distress.

Your team member quickly becomes short of breath, and must sit down. There is no way that will be able to get the scientist out of the forest. You will have to use the resources at your disposal to help your colleague.



All you have is the pack you took with you, which contains a few items for your lunch, as well as some basic scientific materials. You also have a field guide, which may be useful in giving you information about the Asian Giant Hornet.

Your colleague/friend will become unconscious **within 60 minutes**, so work quickly. Fortunately there is an antidote found in the liquid of a common plant to these forests. Unfortunately there is a similar plant with similar liquid that is poisonous when ingested. You must prove which liquid is the Antidote using Test 1-4 on the **Incident Report** to guide you.

Once you have saved your colleague (or haven't...yikes), you have to fill out the **Incident Report** (attached), explaining, in detail, what happened, and what actions you took to help.

Materials and Setup:

1. 200 liquid A
2. 200 mL liquid B
3. 2 pipettes
4. A small square of wax paper (3x3 in)
5. 1 "hints" handout
6. 2 paper towel strips (1 in x 4 in)
7. Salt
8. 2 spoons
9. 2 forks
10. 200 mL of blue Gatorade
11. 4 small containers (150 ml beakers, or Dixie cups) for retrieving both yellow liquids
12. 2 stir rods.

Asian Giant Hornet

Common Name: Asian Giant Hornet

Taxonomy: *Vespa Mandarinia*

Habitat: Various wasp species have been found in much of Eastern Asia, including mountain valleys and forest. Generally, the hornets avoid open plains. It prefers to dig nests in the ground, or using existing nests made by rodents.

Anatomy: Hard exoskeleton, compound eyes, warning coloration, large stinger.

Feeding Ecology: Most adults feed on large insects, and honey from honey bee colonies



Size: Wingspan of 9 centimeters (3 inches). The Asian Giant Hornet is the world's largest hornet, and lives in East Asian. Although they are usually observed flying alone, swarms can be observed if a threat is perceived by the colony. The sting of this wasp can be deadly if an individual is stung enough times, or has developed an allergy. The venom is potent, and has been described to feel like a hot nail is being driven into the skin.

Antidote: There is only one known antidote for a Hornet Wasp's sting. It is in a plant called the Slow-leaf plant (*Pseudotemplo largas*). The Slow-leaf Plant secretes a yellow liquid that can be ingested to counteract the venom of the Hornet Wasp. There is only one problem: The Slow-leaf plant mimics another plant, which is poisonous. In other words, there is another plant that looks almost identical to the Slow-leaf plant, and it is poisonous. It also produces a yellow liquid and the yellow liquid is deadly if ingested.

There are differences between antidote and the liquid made by the poisonous plant. The antidote is a **polar** substance, while the poisonous liquid made by the mimic is **nonpolar**.

It has been said that a few stings from the hornet is enough to be fatal. The antidote should be administered to the victim within a few minutes of being stung, and it is reported that the victim could lose consciousness and die within 60 minutes enduring multiple stings. See below a list of possible **symptoms** from the sting:

1. Anaphylaxis (severe allergic reaction; throat closes)
2. Swelling of site of sting
3. Bruising of site of sting
4. Decrease in heart rate
5. Shallow breathing
6. Victim may become unconscious

Incident Report

Team Members: _____

Date of Incident: _____ Location of Incident: _____

Brief Description of Incident:

Brief Summary of Results: (What were the outcomes to each test you performed? Elaborate.) **You must show the following things in order to prove that you have given the victim the antidote, and not the poison.** FULL SENTENCES MUST BE USED.

Test 1: Using a fork, gently place a paper clip on top of each liquid. What happened to the paperclip in each substance? *Hint: which substance displayed surface tension?

TEST 2: Place approximately 25 mL of each liquid in two separate cups or beakers provided. Mix 3 grams of salt into each liquid. Which substance was able to **dissolve charged/ionic** substances? How do you know?

TEST 3: Using a pipette, place three drops of each liquid on each two separate pieces of wax paper. Which substance displayed **Cohesion**? How do you know?

TEST 4: Using two separate cups or beakers provided, mix approximately 10 mL of each liquid with 10 mL of blue Gatorade. Which substance is able to **mix** with other polar substances? How do you know?

TEST 5: Submerge one end of a paper towel in each liquid. Which substance is able to do Capillary Action? How do you know?

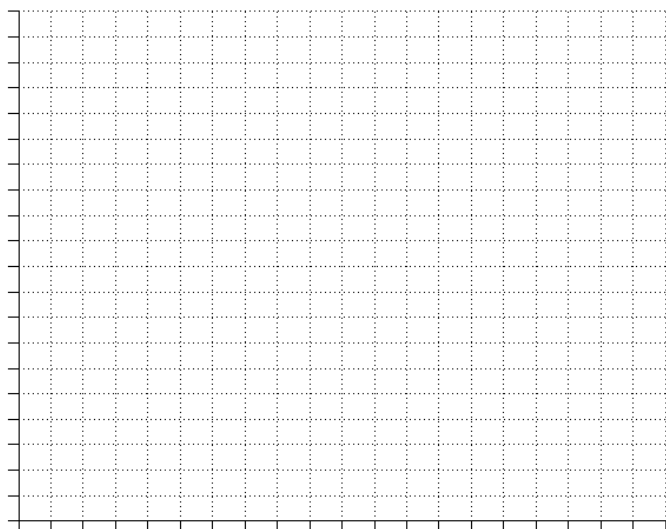
TEST 6: Using the appropriate pipet test how many drops of each liquid will stay on the penny without spilling over. What property of water is this?

TEST 7: Using the treated toothpicks, fill the bottom of a petri dish with water. Sprinkle a small amount of pepper on the water. Insert the treated toothpicks in the center. Explain what happens. In relation to water properties, WHY is this occurring? What do you think is on the treated toothpicks? Now try both liquids, explain your findings.

Data Analysis: Below is some data related to the East Asian Wasp and the plants your team was studying. Plot the data and answer the questions related to the graphs. Make sure to use the appropriate type of graph for each data set. **Your graphs should include:** a title, axis titles with units, and use an appropriate scale.

-Data set 1: Symptoms of Wasp Stings- below is the average % of patients who experienced each symptom of a wasp sting.

Symptom	Swelling at Site of Sting	Bruising at Site of Sting	Bradycardia (decreased HR)	Throat swelling	Shallow Breathing	Hives
% of patients	89%	57%	76%	61%	45%	32%

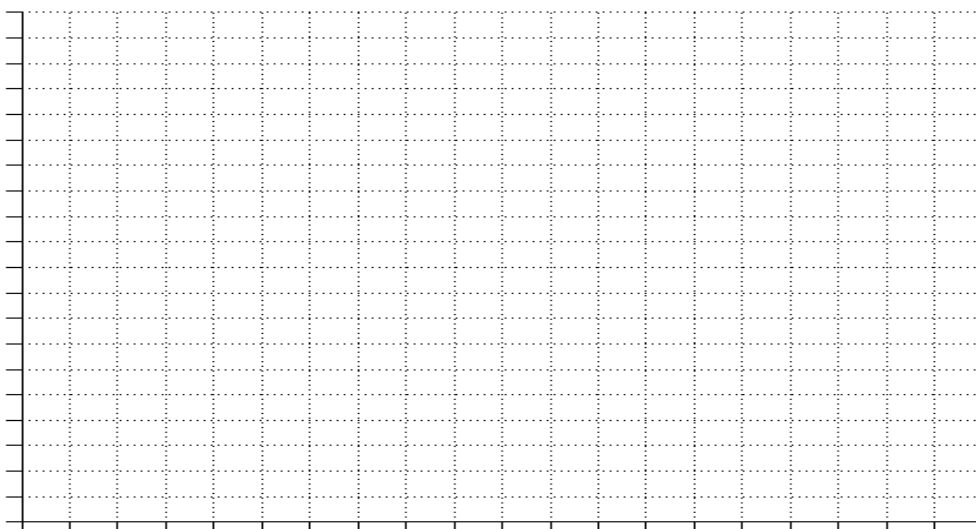


1. What symptoms are patients more likely to develop?

2. What symptoms are patients least likely to develop?

-Data set 2: Heart Rate Changes- below is data on heart rate changes in patients administered the substance A, substance B, and a control patient who was not administered any substance. Heart rate was measured every 5 minutes over the course of 45 minutes. Heart rate is measured in **BPM** (beats per minute)

	5 min	10 min	15 min	20 min	30 min	35 min	40 min	45 min
Substance A	67	60	54	48	42	37	35	32
Substance B	61	68	59	60	64	62	65	67
Control	65	66	68	64	63	60	64	60



1. Why is a control necessary to this data set?
2. What possible conclusions could we draw about this data?

Analysis Questions:

1. What is polarity?
2. How are magnets and water molecules the same?
3. Draw a picture that explains magnetic poles and polarity.

4. Explain why water is a polar molecule.
5. Why is water cohesive?
6. What is the biological importance of water's cohesive and adhesive properties?
7. What is the biological importance of water's relatively high heat of vaporization?
8. Why does ice float? What is the biological importance of ice floating when it freezes?

Conclusion CER

Question: Which substance should we administer to our sick colleague?	
Claim:	
Evidence:	Reasoning:

Action Taken to Treat Patient (Did you administer the antidote in time?):

MURDER AND A MEAL

Pre-Lab Reading and Questions

Why Do We Need Fats, Proteins, and Carbohydrates in Our Diet?

Adapted from *Lions Talk Science*

There are an endless number of diet plans available today that purport to be the answer to all of our weight loss needs. Although these diet systems have led to weight loss for many people, we should be careful about upsetting the balance of the macromolecules we ingest. After all, the body needs all three macromolecules to function properly.

Fats are essential for life because they perform important functions such as insulating us from the cold and providing a cushion for internal organs. Fat in the brain surrounds neurons and allows electrical signals to flow efficiently, giving us the ability to think and act quickly. At the cellular level, fats can act as information carriers and are part of the cell membrane. They are also excellent energy storage molecules. Compared to carbohydrates and proteins, fats contain twice the number of calories per gram and can be stored for years. Fats are found in foods like oils, avocado, dairy products like egg yolks and cheese, meats, and nuts.

Carbohydrates are the body's primary source of energy. Compared to fats they are poor molecules for energy storage and are typically used *quickly* after they are ingested. There are many kinds of complex sugars that can be called "carbohydrates," but the most basic carbohydrate is glucose which is a monosaccharide. Glucose is constantly being converted into ATP, which contains a large amount of energy. There are also disaccharides which are made up of two monosaccharides. Together, mono- and di-saccharides make up a class of carbohydrates called *simple sugars*. These are found in sweeteners, fruit, and table sugar.

Carbohydrates with three or more subunits are called polysaccharides. These make up a class of carbohydrates called *complex carbohydrates*. Simple sugars will give you a small burst of energy, but complex carbohydrates take longer to digest due to their structure, giving you longer lasting energy. These are found in starchy vegetables like potatoes and onions. They are also found in high fiber vegetables, pasta, bread, oatmeal, corn, and rice.

While **proteins** (also called polypeptides) can be used as energy sources, the primary reason we need to ingest proteins is for their nitrogen. Nitrogen is necessary for building new proteins and nucleic acids. Nucleic acids (like DNA) contain our genetic information, which are the blueprints from which proteins are made. The building blocks of proteins are called amino acids. There are 21 amino acids, but our body can only produce 12 of them. The other 9, named essential amino acids, must be ingested in order to make new proteins. Proteins perform all kinds of jobs in the cell including moving molecules, signaling to neighboring cells, and replicating DNA. In the absence of dietary protein, the body will scavenge protein from muscles to produce DNA and protein for more crucial organs of your body

Questions:

1. How do mono-, di-, and polysaccharides differ in *structure*? Give one example of each.
2. How do mono-, di-, and polysaccharides differ in *function*?

3. Why are fats helpful for your cells?
4. Why should you consume proteins?
5. What is the monomer of a polypeptide?

The Case: A murder has occurred right here on our peaceful little town of South Forsyth. As top-notch biology students at SFHS, you have been asked to assist in the investigation of this most unfortunate incident. Central to identifying the individual who committed this crime is establishing where the victim was the day of the crime so that detectives can question the individuals with whom the victim came into contact. An autopsy was performed on the victim and has revealed that the victim ate just prior to the time of death.

Upon questioning the victim's friends and family, detectives working the case have learned that the victim enjoyed eating at the following three restaurants. For each restaurant predict what macromolecules you would **MOST** expect to find in each meal. *Be sure to specify between simple and complex carbohydrates.*

First Watch: The go to brunch place for the victim! The victim would typically order blueberry oatmeal pancakes and a side of bacon!

Prediction:

Taco Mac: The victim would hang out here to watch sporting events while feasting on lemon pepper wings and celery.

Prediction:

Carrabba's: The victim loved to go here for a night of bread, olive oil, and buttery and creamy pasta and finish with strawberry cheesecake.

Prediction:

The forensic pathologist has removed the contents of the victim's stomach for you to analyze in order to determine where the victim had his last meal, and who he had it with. The following suspects submitted the testimonies below.

Suspect #1 = The victim's boss, who claims to have been out that afternoon enjoying brunch at First watch.

Suspect #2 = The victim's roommate and best friend, who claims to have been out with their other buddies at taco mac enjoying wings and watching football.

Suspect #3 = The victim's spouse, who claims to have been out at Carraba's that night.

Testing Procedures: Before analyzing the stomach contents of the deceased, you must determine the procedures used to test for the presence of each macromolecule. Read the information provided below. Use this information to complete table 1 on the next page with the name of the test to be used and what positive and negative results will appear to be.

Lipid Test: Brown Paper Staining

1. Use the pipette to distribute a small amount of sample onto the brown paper
2. Clean any food particles off the surface and allow paper to dry for 10 minutes
3. Observe results.

Negative Result: No translucency to paper

Positive Result: Paper is translucent when held to light

Simple Sugar Test: Benedicts Solution

1. Use a pipette to add 20 drops of sample to a test tube
2. Add 40 drops of Benedicts solution to the test tube
3. Heat in a hot water bath for three minutes
4. Observe the color change

Negative Result: No color change to clear blue solution

Positive Result: the clear blue solution will turn green, yellow, or brick red

Complex Carbohydrate (Starch) Test: Lugol's Iodine Solution

1. Use a pipette to add about 40 drops of sample to the test tube.
2. Add 3-4 drops of iodine
3. Observe the color change

Negative Result: No color change

Positive Result: solution will turn bluish/purple-ish black

Protein Test: Biuret Solution

1. Add 40 drops of sample to the test tube
2. Add 20 drops of Biuret Solution. Cover with thumb and gently swirl to mix.
3. Observe the color change

Negative Result: No color change to blue solution

Positive Result: the blue solution will change color to pink-purple

Part 1: Test Restaurant Samples

Directions: You have been given a sample from each restaurant. Test each sample according to the procedures above. As you test, fill in table 1 on the next page with your observations and if the result was positive or negative for each macromolecule. Observations is what you SEE and result is either POSITIVE or NEGATIVE

Table 1

	Simple Sugar Test	Starch Test	Protein Test	Lipid Test
Meal #1- First Watch	Observations:	Observations:	Observations:	Observations:
	Result:	Result:	Result:	Result:
Meal #2- Taco Mac	Observations:	Observations:	Observations:	Observations:
	Result:	Result:	Result:	Result:
Meal #3- Carraba's	Observations:	Observations:	Observations:	Observations:
	Result:	Result:	Result:	Result:

Part 2: Test the Victim Sample

Directions: Obtain a sample of the stomach contents from the victim from your instructor. Using the same procedures above, test the sample. Record your data in **table 2** below.

Table 2

	Simple Sugar Test	Starch Test	Protein Test	Lipid Test
Victim Sample	Observations:	Observations:	Observations:	Observations:
	Result:	Result:	Result:	Result:

Part 3: Report your results

Directions: Now that you have tested your samples, you need to make a conclusion about the results and communicate it to the police department. Fill out the police report below using your results as well as **prior** knowledge of macromolecules.

Question: Which suspect is most likely the killer?	
Claim:	
Evidence (state what the <i>data shows</i> , do not <i>interpret</i> results)	Reasoning (how does your data prove your claim):

Part 4: Meeting our Nutritional Needs

When considering meeting our own nutritional needs, we must consider what benefits each macromolecule has for our bodies and what our cells use these macromolecules for. Take a look at the nutrition labels for four different meals below:

A	B	C	D
NUTRITION FACTS	NUTRITION FACTS	NUTRITION FACTS	NUTRITION FACTS
Carbohydrates:	Carbohydrates:	Carbohydrates:	Carbohydrates:
Simple- 8g	Simple- 6g	Simple- 0g	Simple- 7g
Complex-4g	Complex-2g	Complex-0g	Complex-4g
Lipids- 7g	Lipids- 0g	Lipids- 9g	Lipids- 5g
Proteins-0g	Proteins-5g	Proteins-7g	Proteins-6g

1. If you started a weightlifting routine and wanted to build muscle, which meal above would be the most beneficial and why?
2. Which meal would a runner want to eat right before running a marathon? Why?
3. Which meal would be most beneficial to an animal that lives in very cold environments and why?
4. Which meal above would you choose on a daily basis and why?